

IMPERIAL COLLEGE LONDON

M.Eng EXAMINATION IN CHEMICAL ENGINEERING 2018

PART II

For Internal Students of Imperial College London

This paper is also taken for the relevant examination
for the Associateship

TRANSFER PROCESSES 2

Monday 4th June 2018: 14:00-16:30

Answer **Four** questions

Two from section I, one from section II and one from section III

All questions carry equal marks

A psychrometric chart and formula sheet are provided separately

**THIS EXAMINATION PAPER HAS EIGHT PAGES IN TOTAL
WHICH INCLUDES THIS COVER SHEET**

TURN OVER FOR QUESTIONS

SECTION I

Question 1

[25 marks]

- (a) Pure, saturated and quiescent water vapour condenses on the outer surface of a vertical tube which has an outer diameter of 60 mm and is 0.5 m long. The outer wall temperature is kept at 8 K below the saturation temperature, and the condensate film Reynolds number at the bottom of the tube is 80. The specific latent heat of vaporisation is $2257 \text{ kJ kg}^{-1} \text{ K}^{-1}$, and the condensate liquid viscosity is $2.79 \times 10^{-4} \text{ Pa s}$. Calculate the average heat transfer coefficient along the tube.

[10 marks]

- (b) If the tube described in part (a) is placed horizontally, how would the average heat transfer coefficient change and by what percentage?

[4 marks]

- (c) Consider condensation of water vapour on a flat, vertical surface. The vapour is pure, saturated and flows down the surface parallel to the condensate film. At a certain location along the surface, the interfacial shear stress is 4.0 Pa and the mean condensate film velocity is 0.6 m s^{-1} . The condensate liquid density is 958 kg m^{-3} , thermal conductivity is $0.68 \text{ W m}^{-1} \text{ K}^{-1}$, and viscosity is $2.79 \times 10^{-4} \text{ Pa s}$. Assuming shear-controlled laminar film condensation, calculate the local film thickness and the heat transfer coefficient.

[11 marks]

SECTION I

Question 2

[25 marks]

- (a) An air stream with a humidity of 0.003 kg kg^{-1} (dry air) has a dry-bulb temperature of 21°C . Use the provided psychrometric chart to determine the relative humidity, the wet-bulb temperature, and the specific enthalpy of the air stream.

[3 marks]

- (b) The above air stream enters a steam-spray humidifier at an equivalent dry air flow rate of 2 kg s^{-1} . Saturated steam having a specific enthalpy of 2676 kJ kg^{-1} is injected into the air stream so that the air leaves the humidifier with a relative humidity of 50%. You can assume that there is no heat transfer with the surroundings, and the pressure is constant throughout.

(i) What is the required steam flow rate?

[10 marks]

(ii) What is the temperature of the air stream at the exit?

[2 marks]

- (c) The Merkel method was developed by considering mass and energy balances in a forced-draught counter-flow cooling tower packing unit. Describe, with the help of diagrams, how the Merkel integral varies with the water to air flow rate ratio.

[10 marks]

SECTION II

Question 3

[25 marks]

- (a) Define “mean beam length”, and calculate the mean beam length for a cube of side length 3 m.

[4 marks]

- (b) A grey sphere with emissivity of 0.8 and diameter of 0.1 m is surrounded by an enclosure with a surface area of 0.4 m^2 and emissivity 0.6. The sphere is electrically heated, and its temperature is maintained at 900 K with a power input of 800 W. Both the sphere and the enclosure are grey and opaque, and radiation is the only mechanism of heat transfer. The Stefan-Boltzmann constant is $5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$.

- (i) Write down the overall heat balance equation for the sphere.

[5 marks]

- (ii) Calculate the steady state temperature of the enclosure.

[16 marks]

SECTION II

Question 4

[25 marks]

- (a) Define “view factor”, and calculate all the view factors between a sphere and an infinitely large plane, assuming the sphere is held close to the plane and its surface area is negligible compared to that of the plane.

[6 marks]

- (b) Two identical, flat, horizontal plates are aligned parallel to each other. The width of each plate is 2 m, while the distance between the plates is 1 m. The length of the plates is much larger compared to the other dimensions, so that changes in the direction of length can be ignored. As a result, the view factors between the plates can be estimated using the crossed-string method for two-dimensional configurations. Both plates are grey and opaque with emissivity of 0.6. The temperature of the upper plate is T_1 , while that of the lower plate is T_2 . You may assume that the surroundings are black and cold, and radiation is the only mode of heat transfer. The Stefan-Boltzmann constant is $5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$.

- (i) Derive an expression for the net radiative flux from the upper plate.

[12 marks]

- (ii) If the upper plate is held at a constant temperature of 1000 K, calculate the temperature of the lower plate.

[7 marks]

SECTION III

Question 5

[25 marks]

Consider steady unidirectional, incompressible flow between two parallel plates, which are separated by a distance, H , as shown in Figure 1. The gap between the two plates is filled with two fluids, namely oil (on top, with density ρ_o and viscosity, μ_o) and water (with density $\rho_w > \rho_o$ and viscosity, μ_w), with an interface at $y = H/2$. The lower plate is stationary, while the upper plate moves at a constant speed, U_0 . Gravity acts vertically downward and the flow is fully developed. Use two-dimensional Cartesian coordinates (x, y) with corresponding velocity components (u, v) and pressure p .

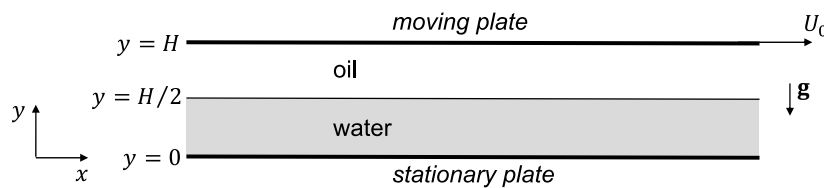


Figure 1 – Geometry for the flow between two parallel plates.

- (a) Verify that the velocity component v is zero at any (x, y) location.

[3 marks]

- (b) Simplify the x - and y -component momentum equations, clearly stating the relevant assumption for each term that can be dropped.

[6 marks]

- (c) Solve for the u velocity component in each layer. You are given that the pressure gradient is constant and equal to zero across both fluids.

[7 marks]

- (d) Determine the pressure profile between the two parallel plates. You may assume $p = p_0$ at $y = 0$.

[2 marks]

- (e) Sketch the velocity and stress profiles in the gap between the plates for the following two situations (i) $\mu_o = 3\mu_w$ and (ii) $\mu_o = \mu_w$.

[7 marks]

SECTION III

Question 6

[25 marks]

Consider flow through an annular die to form a tube polymer, as shown in Figure 2. In the die of length L the liquid polymer of viscosity μ flows in the annular region between an inner solid cylinder and a hollow outer cylinder; their radii are R_2 and R_1 , respectively. The flow is assumed to be steady and unidirectional, with the only non-zero velocity component being the z -component, u_z . The latter is also independent of the angular location, θ . Gravity can be neglected.

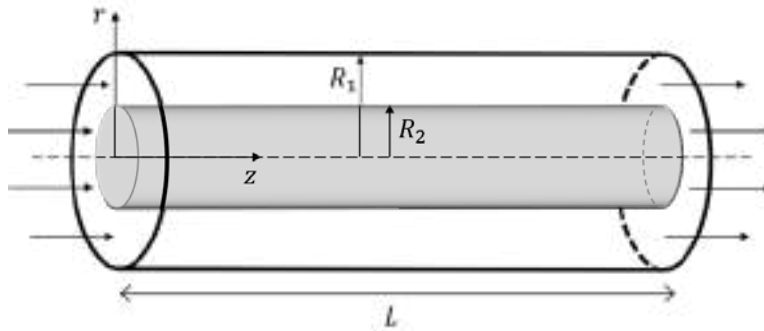


Figure 2 – Geometry for the flow through an annular die.

- (a) Verify that u_z is independent of distance, z .

[3 marks]

- (b) Simplify the z -component momentum equation, clearly stating the relevant assumption for each term that can be dropped.

[4 marks]

- (c) Determine the velocity profile, $u_z(r)$. You may assume that the pressure gradient, $\partial p / \partial z$, is constant and independent of r .

[7 marks]

- (d) Determine the location of maximum velocity within the annulus. Assume $R_1 = 10$ cm and $R_2 = 5$ cm.

[4 marks]

QUESTION 6 CONTINUES

TURN OVER FOR Parts (e) and (f)

- (e) Demonstrate why the maximum velocity is *not* located at the halfway point between the inner and outer tubes.

[5 marks]

- (f) Sketch the velocity and stress profiles in the annulus.

[2 marks]

END OF QUESTION 6

IMPERIAL COLLEGE LONDON

M.Eng EXAMINATION IN CHEMICAL ENGINEERING 2019

PART II

For Internal Students of Imperial College London

This paper is also taken for the relevant examination
for the Associateship

TRANSFER PROCESSES 2

Tuesday 4th June 2019: 10:00-12:30

Answer **ALL** four questions

All questions carry equal marks

**Use yellow answer booklets for Question 1,
and green booklets for Questions 2-4.**

**THIS EXAMINATION PAPER HAS SEVEN PAGES IN TOTAL
WHICH INCLUDES THIS COVER SHEET**

TURN OVER FOR QUESTIONS

Question 1**[25 marks]**

Consider steady unidirectional, incompressible flow in a vertical slit formed by two parallel walls with dimensions $H \times L$ (vertical $\times$ transverse direction), which are separated by a distance, D , as shown in Figure 1. The flow is sustained by an externally applied pressure gradient ($\partial p / \partial z > 0$) in the z -direction, and by the vertical motion of the left-hand plate at speed $W_0 > 0$ in the z -direction. Gravity acts vertically downward and the flow is fully developed. L is much longer than H and D , and edge effects can therefore be neglected. Note that the Cartesian coordinate system has origin at the midplane of the slit ($x = 0$).

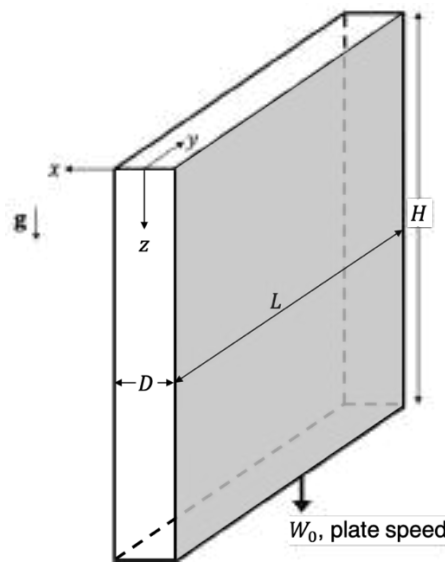


Figure 1 – Geometry for the flow in a vertical slit with one plate moving.

- (a) Verify that the velocity component $w = w(x)$ and that $p = p(z)$.

[4 marks]

- (b) Simplify the z -component of the momentum equation, clearly stating the relevant assumption for each term that can be dropped.

[5 marks]

- (c) Solve for the w velocity component. You are given that the term, $\partial p / \partial z - \rho g_z$, is constant and larger than zero.

[5 marks]**QUESTION 1 CONTINUES**

(d) Determine the average velocity of the fluid through the slit.

[2 marks]

(e) Determine the position of the maximum velocity within the slit. How does the result compare to the situation where $W_0 = 0$?

[2 marks]

(f) Sketch the stress and velocity profiles in the gap between the plates for the following two situations:

(i) $W_0 > 0$

(ii) $W_0 = 0$.

[7 marks]

END OF QUESTION 1

Question 2**[25 marks]**

In a kettle-type reboiler, a static pool of water is heated by a bundle of stainless steel tubes of external diameter 5 cm submerged in water. The water in the reboiler is saturated at 1 atm and the vapour formed at the surface of the pool is fed to a distillation column.

- (a) Consider pool boiling of the water in the reboiler:
- (i) Draw the boiling curve and identify the different boiling regimes.
 - (ii) Describe the heat transfer mechanisms in each regime.
 - (iii) Why is the burnout point (critical heat flux) avoided in the design of boilers?

[5 marks]

- (b) The total length of the stainless steel tubes in the bundle is 30 m and their surface is maintained at a temperature of 109 °C. The surface correction factor for chemically etched stainless steel is $C_{sf} = 0.0133$ and the exponent is $s = 1.0$.
- (i) Calculate the power required to cause boiling.
 - (ii) Assuming steady state, calculate the mass flow rate of vapour to the distillation column.

[8 marks]

- (c) Calculate the total heat flux for a heater surface temperature of 350 °C and a surface emissivity of 0.75. The Stefan-Boltzmann constant is $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$.

[7 marks]

- (d) Consider flow boiling in vertical pipes with upward flow and uniform heat flux.
- (i) Name the main flow boiling regimes in the order in which they occur in a vertical tube.
 - (ii) What is the dominant heat transfer mechanism in each region?
 - (iii) How does the heat transfer coefficient vary as a function of distance along the pipe? Sketch the qualitative trend.

[5 marks]

**TURN OVER FOR
PROPERTY TABLE FOR QUESTION 2**

Property Table for Question 2

Symbols:

μ : dynamic viscosity

ρ : density

ν : kinematic viscosity

c_p : heat capacity at constant pressure

h_{lv} : latent heat of vaporisation

σ : interfacial tension

k : heat conductivity

Liquid properties:

T (°C)	μ_l (Pa s)	ρ_l (kg m ⁻³)	$c_{p,l}$ (J kg ⁻¹ K ⁻¹)	σ (N m ⁻¹)	k_l (W m ⁻¹ K ⁻¹)
100	0.279×10^{-3}	957.9	4217	58.9×10^{-3}	0.68
160	0.173×10^{-3}	909.9	4331	47.2×10^{-3}	0.67
220	0.124×10^{-3}	844.6	4590	33.9×10^{-3}	0.64

Vapour properties:

T (°C)	h_{lv} (J kg ⁻¹)	ρ_v (kg m ⁻³)	ν_v (m ² s ⁻¹)	$c_{p,v}$ (J kg ⁻¹ K ⁻¹)	k_v (W m ⁻¹ K ⁻¹)
100	2.257×10^6	0.5956	0.201×10^{-4}	2029	0.0248
160	2.091×10^6	3.31	0.241×10^{-4}	1996	0.0268
220	1.870×10^6	10.84	0.398×10^{-4}	1983	0.0358

END OF QUESTION 2

Question 3**[25 marks]**

In a water-tube boiler, water is heated within N pipes inside a furnace. The only mechanism of heat transfer between the furnace and the pipes is by radiation. The wall of the furnace is black and is held at constant temperature T_f . The pipes are grey and opaque with emissivity $\varepsilon_p = 0.5$.

- (a) Assuming that all N pipes are at the same temperature, T_p , and that they can be treated as one body, show that the total radiation heat flux leaving the pipes is:

$$W_p = \frac{\sigma(T_p^4 + T_f^4 F_{pf})}{1 + F_{pf}}$$

where σ is the Stefan-Boltzmann constant and F_{pf} is the view factor from the pipes to the furnace walls.

[15 marks]

- (b) For the case of one pipe ($N = 1$), and assuming that the pipe and furnace are arranged as concentric cylinders, calculate the temperature of the surface of the pipe for the following conditions: the water flowing inside the pipe removes heat at a rate per unit length of the pipe of 8 kW m^{-1} ; the radius of the pipe is 0.1 m and the radius of the furnace is 1 m ; the temperature of the wall of the furnace is 1000 K . You are given that the view factor from cylinder “2” (outer cylinder with radius r_2) to cylinder “1” (inner cylinder with radius r_1) is: $F_{21} = r_1/r_2$. The Stefan-Boltzmann constant is $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$.

[7 marks]

- (c) What is the colour of a black body?

[3 marks]**END OF QUESTION 3**

Question 4**[25 marks]**

- (a) Describe the different modes of condensation and how they affect the heat transfer coefficient.

[3 marks]

- (b) Consider laminar film condensation on the outer surface of a vertical cylinder. The condensing vapour is pure and saturated. The height of the cylinder is L and the distance from the top of the cylinder is x . The outer diameter of the cylinder is D , and it is greater than the film thickness δ , $D \gg \delta$, for all values of $\delta = \delta(x)$.
- (i) When is laminar film condensation considered to be gravity-controlled and when shear-controlled? Clearly state the relevant assumptions in each case.
- (ii) For gravity-controlled laminar film condensation, apply conservation of energy to a differential element of condensate film of length dx to show that the film thickness varies as

$$\delta(x) = \left[\frac{4k_l \mu_l (T_{sat} - T_w) x}{\rho_l (\rho_l - \rho_v) g h_{lv}} \right]^{1/4}.$$

You are given the dependence of the condensate mass flow rate per unit width, Γ , on the film thickness, δ :

$$\Gamma(\delta) = \frac{\rho_l (\rho_l - \rho_v) g \delta^3}{3\mu_l}.$$

Here, ρ_l and ρ_v are the liquid and vapour densities, respectively, μ_l is the liquid viscosity and g is the acceleration due to gravity.

[15 marks]

- (c) Describe the effect of a non-condensable gas on the condensation of a vapour on a plate, and outline the steps to calculate the total heat flux from the vapour to the plate in this case.

[7 marks]**END OF QUESTION 4**

IMPERIAL COLLEGE LONDON

M.Eng EXAMINATION IN CHEMICAL ENGINEERING 2020

PART II

For Internal Students of Imperial College London

This paper is also taken for the relevant examination
for the Associateship

TRANSFER PROCESSES 2

Monday 18th May 2020 14:00-16:30

Answer **ALL** four questions

All questions carry equal marks

**THIS EXAMINATION PAPER HAS FIVE PAGES IN TOTAL
WHICH INCLUDES THIS COVER SHEET**

TURN OVER FOR QUESTIONS

Question 1**[25 marks]**

Consider steady, unidirectional, incompressible flow in a pipe of radius R and length L , as shown in Figure 1. The pipe is filled with two immiscible liquids and the flow is stratified, meaning that liquid 1 (density, ρ_1 and viscosity, μ_1) flows around liquid 2 (density, ρ_2 and viscosity, μ_2) with an interface at $r = R/2$. The flow is fully developed and the effects of gravity can be considered negligible. You may further assume that the pressure gradient, $-dp/dz = (P_1 - P_2)/L > 0$, is constant and equal in both fluids.

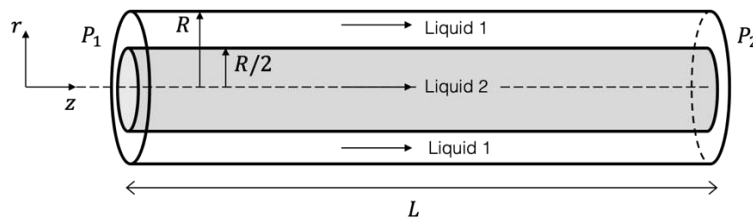


Figure 1 – Geometry for the flow of two immiscible fluids in a cylindrical pipe.

- (a) Using cylindrical coordinates (r, θ, z) and assuming axisymmetry, write and simplify the momentum equations in the z direction for both liquids. Clearly state the relevant assumption for each term that can be dropped.

[6 marks]

- (b) Provide expressions for the boundary conditions that apply to both fluids.

[4 marks]

- (c) Solve for the u_z velocity component in each liquid layer.

[6 marks]

- (d) Obtain expressions for the shear stress, τ_{rz} , in each liquid layer.

[2 marks]

- (e) Sketch the velocity and stress profile in the pipe for the following two situations:

(i) $\mu_1 > \mu_2$ and (ii) $\mu_2 > \mu_1$

[7 marks]**END OF QUESTION 1**

Question 2

[25 marks]

- (a) Air at 1 atm, 34°C and 60% relative humidity needs to be brought to the "comfort" conditions of 24°C and 50% relative humidity using cooling and heating coils. The mass flow rate of the dry air is 100 kg h⁻¹.

- (i) Draw a sketch of the required air-conditioning process in a simplified psychrometric chart.

[5 marks]

Using the provided psychrometric chart, estimate the following amounts:

- (ii) The condensate mass flow rate collected.

[5 marks]

- (iii) The heat rate in the cooling coil.

[5 marks]

- (iv) The heat rate in the heating coil.

[5 marks]

- (b) Inside a duct, warm water at 40°C and cold atmospheric air at 20°C and 20% relative humidity are encountered in counterflow mode, as shown in Figure 2.

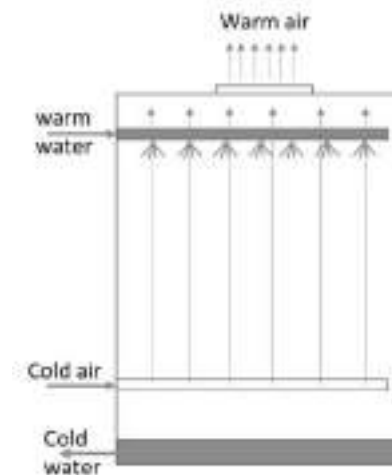


Figure 2 – Schematic of the duct with warm water and cold air in counterflow.

Using the provided psychrometric chart, estimate the minimum temperature that the cold water could achieve at the exit.

[5 marks]

END OF QUESTION 2

Question 3**[25 marks]**

- (a) Steam at 0.27 bar condenses on the surface of 10 horizontal tubes of 50 mm outer diameter and 2 m length arranged in a vertical tier. The tubes surface temperature is 37°C and is constant. Using the available equations in the Equation Sheet and the provided data, calculate:

- (i) The rate of heat transfer.

[10 marks]

- (ii) The rate of condensation.

[5 marks]

- (iii) Explain any key assumptions made to obtain your solution.

[5 marks]

Note: For this vertical tier of tubes, you can approximate the condensate film Reynolds number to $Re_\delta = 2\Gamma/\mu_l$, where Γ is the condensate flow per unit width

Data for Question 3:

Saturation temperature	T_{sat}	340	K
Liquid Density	ρ_l	987.2	kg m ⁻³
Vapour Density	ρ_v	0.174	kg m ⁻³
Latent heat of vaporisation	h_{lv}	2342	kJ kg ⁻¹
Specific heat capacity	c_{pl}	4.182	kJ kg ⁻¹ K ⁻¹
Liquid viscosity	μ_l	577×10^{-6}	N s m ⁻²
Liquid thermal conductivity	k_l	0.640	W m ⁻¹ K ⁻¹
Gravity	g	9.81	m s ⁻²

- (b) Consider laminar film condensation on a vertical flat plate, which is kept at a constant temperature T_w . The condensing vapour is saturated at T_{sat} . Assume that the Nusselt theory applies.

- (i) Estimate the heat flux at the bottom of the plate. You can assume that the film thickness at this location is 0.05 mm, and that the temperature difference between the wall and the vapour is 2 °C. The liquid thermal conductivity is 0.64 W m⁻¹ K⁻¹.

[5 marks]**END OF QUESTION 3**

Question 4**[25 marks]**

Consider two concentric spheres with diameter D_1 and D_2 , and temperatures T_1 and T_2 , respectively. The diameter of sphere 1 is the double that of sphere 2 ($D_1 = 2D_2$), as shown in Figure 3. The surfaces of sphere 2 and sphere 1 act as heat source and heat sink, respectively. Both surfaces have a uniform temperature and radiation is the only mechanism of heat transfer.

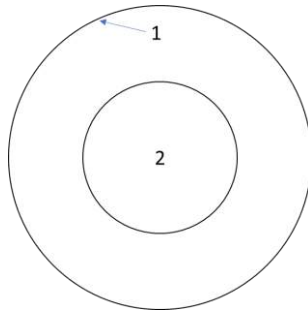


Figure 3. Concentric spheres with $D_1 = 2D_2$

- (a) Find the view factors between the two surfaces.

[5 marks]

- (b) Assume that the spheres are black and opaque. Find an expression for the net rate of radiation between the two surfaces.

[5 marks]

- (c) Assume that the spheres are grey, opaque, and with emissivity ϵ_1 and ϵ_2 , respectively. Find expressions for the fluxes leaving each of the surfaces (that is, W_1 and W_2) and calculate their values for the following conditions:

Data for Question 4c:

Emissivity	ϵ_1	0.8	
Emissivity	ϵ_2	0.8	
Stefan-Boltzmann constant	σ	5.67×10^{-8}	$\text{W m}^{-2} \text{K}^{-4}$
Surface temperature of sphere 1	T_1	300	K
Surface temperature of sphere 2	T_2	500	K

[15 marks]**END OF PAPER**

IMPERIAL COLLEGE LONDON

M.Eng EXAMINATION IN CHEMICAL ENGINEERING 2021

PART II

For Internal Students of Imperial College London

This paper is also taken for the relevant examination
for the Associateship

TRANSFER PROCESSES 2

Monday 10th May 2021: 10.00 – 12:30

Answer **ALL** questions

Question 1 carries 35 marks

Question 2 carries 22 marks

Question 3 carries 23 marks

Question 4 carries 20 marks

**THIS EXAMINATION PAPER HAS SIX PAGES IN TOTAL
WHICH INCLUDES THIS COVER SHEET**

TURN OVER FOR QUESTIONS

Question 1**[35 marks]**

Consider the parallel-plate rotational rheometer depicted in Figure 1. The instrument consists of two horizontal parallel disks separated by a distance h . The gap between the two disks is filled with a liquid of density ρ . The lower disk is stationary, while the upper disk rotates with angular velocity $\Omega = u_\theta r$, thereby generating a steady unidirectional flow in the liquid with velocity $u_\theta = u_\theta(r, z)$ in appropriate cylindrical coordinates, (r, θ, z) . You may assume that the fluid pressure depends on the axial position z only. Gravity acts vertically downward, $\mathbf{g} = [0, 0, -g]$.

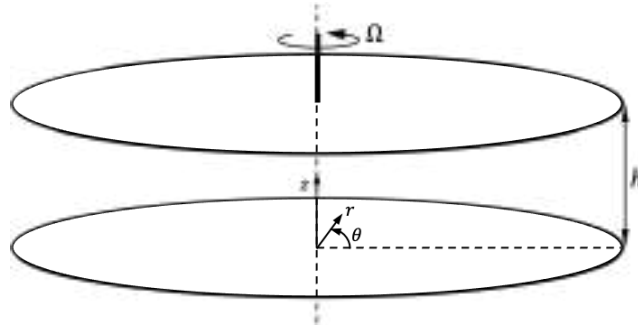


Figure 1. Parallel-plate rotational rheometer.

- (a) Write and simplify the momentum equations in the z and θ directions. Clearly state the relevant assumption for each term that can be dropped.

[10 marks]

- (b) Solve for the $u_\theta = u_\theta(r, z)$ velocity component by assuming that $u_\theta = rf(z)$.

[10 marks]

- (c) Determine the pressure distribution in the fluid.

[2 marks]

- (d) Obtain an expression for the relevant shear stress component in the liquid.

[4 marks]

- (e) Sketch the velocity and shear stress vector fields within the parallel-plate rotational rheometer.

[9 marks]**END OF QUESTION 1**

Question 2**[22 marks]**

- (a) Consider pool boiling and a resulting isolated spherical bubble in the bulk of liquid. Assume that the bubble also contains a non-condensable gas. For this bubble to remain intact (i.e. neither grow nor collapse), it must be at mechanical and thermal equilibrium with the surrounding liquid.

- (i) Derive a mechanical equilibrium equation – that is a force balance – for the suspended bubble. Define all parameters used in the derivation.

[5 marks]

- (ii) Consider the Clausius-Clapeyron equation from thermodynamics,

$$\frac{dP}{dT} = \frac{h_{lv}}{T_{sat}(v_v - v_l)}$$

where h_{lv} is the latent heat of vaporization and v_v and v_l the specific volume of the vapour and liquid phase, respectively. Infer the integral equilibrium relation between bubble size, the amount of superheat and amount of non-condensable gas.

[5 marks]

- (iii) Discuss how this bubble will grow or collapse with the amount of superheat and the effect of the non-condensable gas.

[5 marks]

- (b) In an experiment on pool boiling, a nickel wire working as a heating element is immersed horizontally in water at 1.52 bar pressure. The wire has a length of 20 cm and a diameter of 1 cm. Using a power supply, the current intensity is increased in the wire and at 200 ampere the nickel wire burns out. Using the thermophysical properties of saturated water provided in Table 1 (on page 5), calculate the voltage at burnout.

[7 marks]**END OF QUESTION 2**

Question 3**[23 marks]**

(a) Consider film condensation of a saturated steam at 0.75 bar on a vertical tube, which is kept at a constant temperature $T_w = 345$ K. The tube is 45 cm long and 60 mm wide.

(i) Calculate the rate of heat transfer and explain any key assumption to obtain your solution. You may use the thermophysical properties of saturated water provided in Table 1 (on page 5).

[8 marks]

(ii) Calculate the total condensation mass rate per hour.

[3 marks]

(iii) Consider a second longer vertical tube of the same diameter. Estimate the tube height necessary for the condensate flow just to become turbulent,

[5 marks]

Note: For this vertical tube, you can approximate the condensate film Reynolds number to $Re_\delta = 4\Gamma/\mu_l$, where Γ is the condensate mass flow rate per unit width and μ_l is the liquid viscosity.

(b) Consider now the same film condensation process, but on a bank of 80 horizontal tubes, 8 in each vertical column and 10 in each horizontal row. The tubes are all 45 cm long and 60 mm wide.

(i) Derive an equation that relates the average heat transfer coefficient in a vertical tube to the average heat transfer in a vertical column of horizontal tubes.

[3 marks]

(ii) Calculate the total rate of condensation collected per hour on the bank of tubes.

[4 marks]**END OF QUESTION 3**

Table 1. Thermophysical properties of saturated water

Temperature, T (K)	Pressure, p (bar) ^a	Specific Volume (m ³ /kg)		Heat of Vapor- ization, h_{fg} (kJ/kg)	Specific Heat (kJ/kg · K)		Viscosity (N · s/m ²)		Thermal Conductivity (W/m · K)		Prandtl Number		Surface Tension, $\sigma_f \cdot 10^3$ (N/m)
		$v_f \cdot 10^2$	v_g		$c_{p,f}$	$c_{p,g}$	$\mu_f \cdot 10^4$	$\mu_g \cdot 10^4$	$k_f \cdot 10^2$	$k_g \cdot 10^2$	Pr_f	Pr_g	
273.15	0.00611	1.000	206.3	2502	4.217	1.854	1750	8.02	569	18.2	12.99	0.815	75.5
275	0.00697	1.000	181.7	2497	4.211	1.855	1652	8.09	574	18.3	12.22	0.817	75.3
280	0.00990	1.000	130.4	2485	4.198	1.858	1422	8.29	582	18.6	10.26	0.825	74.8
285	0.01387	1.000	99.4	2473	4.189	1.861	1225	8.49	590	18.9	8.81	0.833	74.3
290	0.01917	1.001	69.7	2461	4.184	1.864	1080	8.69	598	19.3	7.56	0.841	73.7
295	0.02617	1.002	51.94	2449	4.181	1.868	959	8.89	606	19.5	6.62	0.849	72.7
300	0.03531	1.003	39.13	2438	4.179	1.872	855	9.09	613	19.6	5.83	0.857	71.7
305	0.04712	1.005	29.74	2426	4.178	1.877	769	9.29	620	20.1	5.20	0.865	70.9
310	0.06221	1.007	22.93	2414	4.178	1.882	695	9.49	628	20.4	4.62	0.873	70.0
315	0.08132	1.009	17.82	2402	4.179	1.888	631	9.69	634	20.7	4.16	0.883	69.2
320	0.1053	1.011	13.98	2390	4.180	1.895	577	9.89	640	21.0	3.77	0.894	68.3
325	0.1351	1.013	11.06	2378	4.182	1.903	528	10.09	645	21.3	3.42	0.901	67.5
330	0.1719	1.016	8.82	2366	4.184	1.911	489	10.29	650	21.7	3.15	0.908	66.6
335	0.2167	1.018	7.09	2354	4.186	1.920	453	10.49	656	22.0	2.88	0.916	65.8
340	0.2713	1.021	5.74	2342	4.188	1.930	420	10.69	660	22.3	2.66	0.925	64.9
345	0.3372	1.024	4.683	2329	4.191	1.941	389	10.89	664	22.6	2.45	0.933	64.1
350	0.4163	1.027	3.846	2317	4.195	1.954	365	11.09	668	23.0	2.29	0.942	63.2
355	0.5100	1.030	3.180	2304	4.199	1.968	343	11.29	671	23.3	2.14	0.951	62.3
360	0.6209	1.034	2.645	2291	4.203	1.983	324	11.49	674	23.7	2.02	0.960	61.4
365	0.7514	1.038	2.212	2278	4.209	1.999	306	11.69	677	24.1	1.91	0.969	60.5
370	0.9040	1.041	1.861	2265	4.214	2.017	289	11.89	679	24.5	1.80	0.978	59.5
373.15	1.0133	1.044	1.679	2257	4.217	2.029	279	12.02	680	24.8	1.76	0.984	58.9
375	1.0815	1.045	1.574	2252	4.220	2.036	274	12.09	681	24.9	1.70	0.987	58.6
380	1.2869	1.049	1.337	2239	4.226	2.057	260	12.29	683	25.4	1.61	0.999	57.6
385	1.5233	1.053	1.142	2225	4.232	2.080	248	12.49	685	25.8	1.53	1.004	56.6

Question 4**[20 marks]**

(a) An air stream is at 35°C dry-bulb temperature, 21°C wet-bulb temperature and 1 atm. Using the provided psychrometric chart, estimate the following amounts:

(i) The humidity, the relative humidity and the specific enthalpy of the air stream.

[5 marks]

(b) 10 kg s⁻¹ of the air stream is cooled with a spray of cold water. The resulting stream has a relative humidity of 90%. Using the provided psychrometric chart, estimate the following amounts:

(i) Dry-bulb and wet-bulb temperatures in the resulting stream. Explain any key assumptions.

[5 marks]

(ii) The rate at which water vaporises.

[5 marks]

(c) The cooled air is then mixed with 20 kg s⁻¹ of warm air with a relative humidity of 10% and a dry-bulb temperature of 40°C. What is the resultant dry-bulb temperature?

[5 marks]**END OF QUESTION 4**